

Report of optical and XUV properties of LiF

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Simulations parameters.

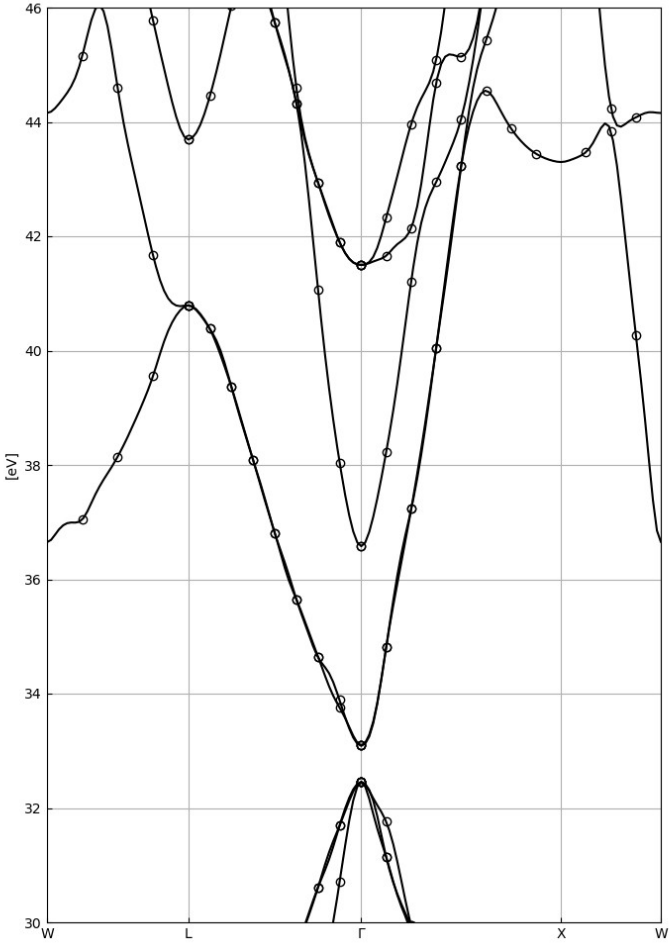
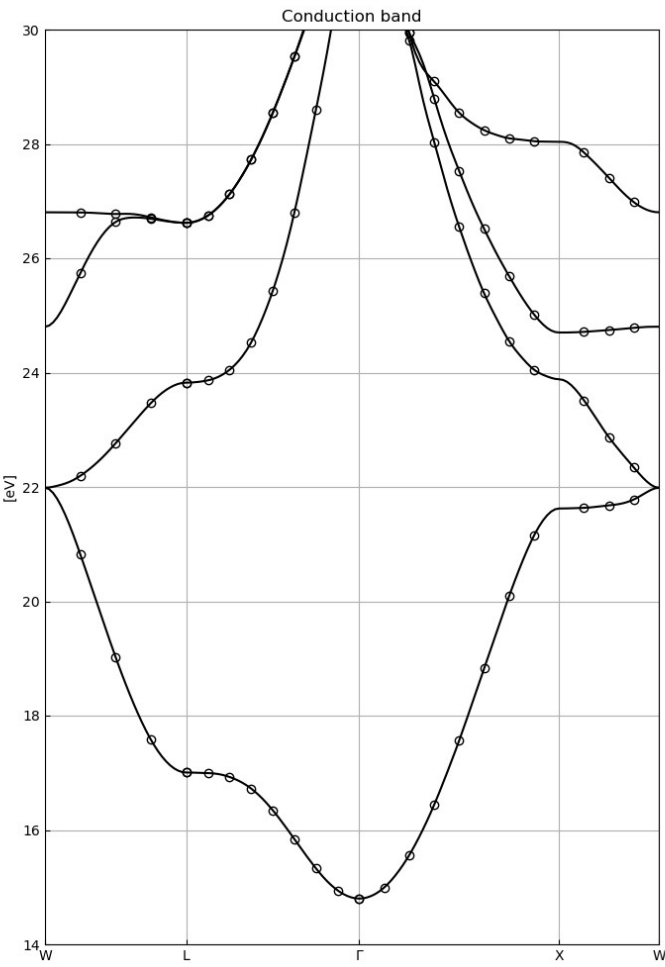
- SCF 6x6x6, ecut 80 Ry
- NSCF. 2 kpts grids 16x16x16, and 8x8x8. (100 bands in total)
- Screening 59 RL, 50 Bands

Following: some plots of the band structure

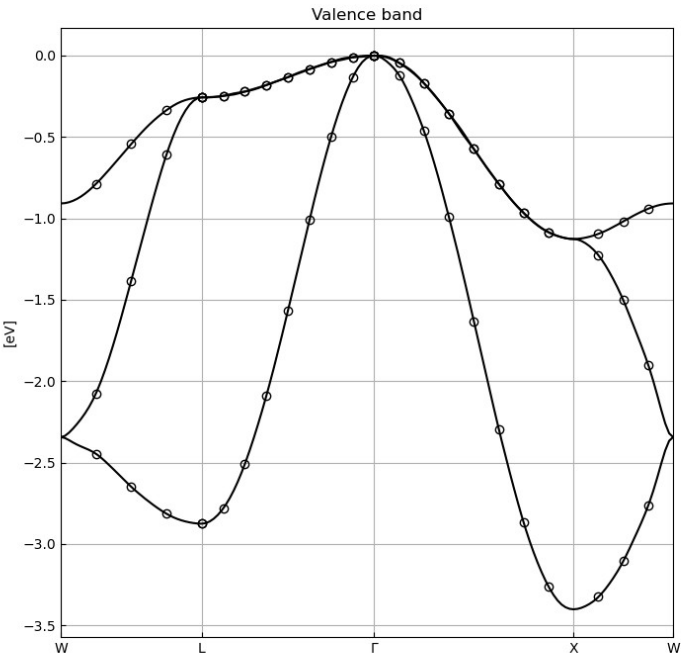
Scissor operator of 6 eV, plus a stretching of 5%.

BSE simulations. Band structure, 6 eV scissor and 5% (20%) stretching. The absolute position of the peaks should be not taken as exact, especially for higher energies. Indeed, even when QP corrections are computed, usually theory and experiment are aligned (especially for X-edges but sometimes also for the visible/UV spectra). Moreover the broadening is not computed, but it is arbitrarily set to 0.1 eV for all peaks.

Below: Conduction band up to 30 eV

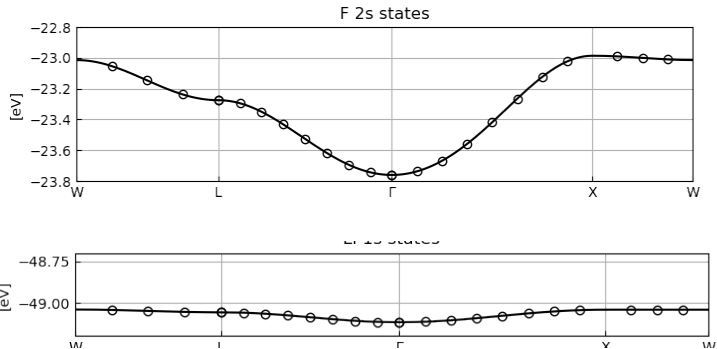


Above: Conduction band from 30 to 46 eV



Above: valence band

Below F(2s) and Li(1s) states

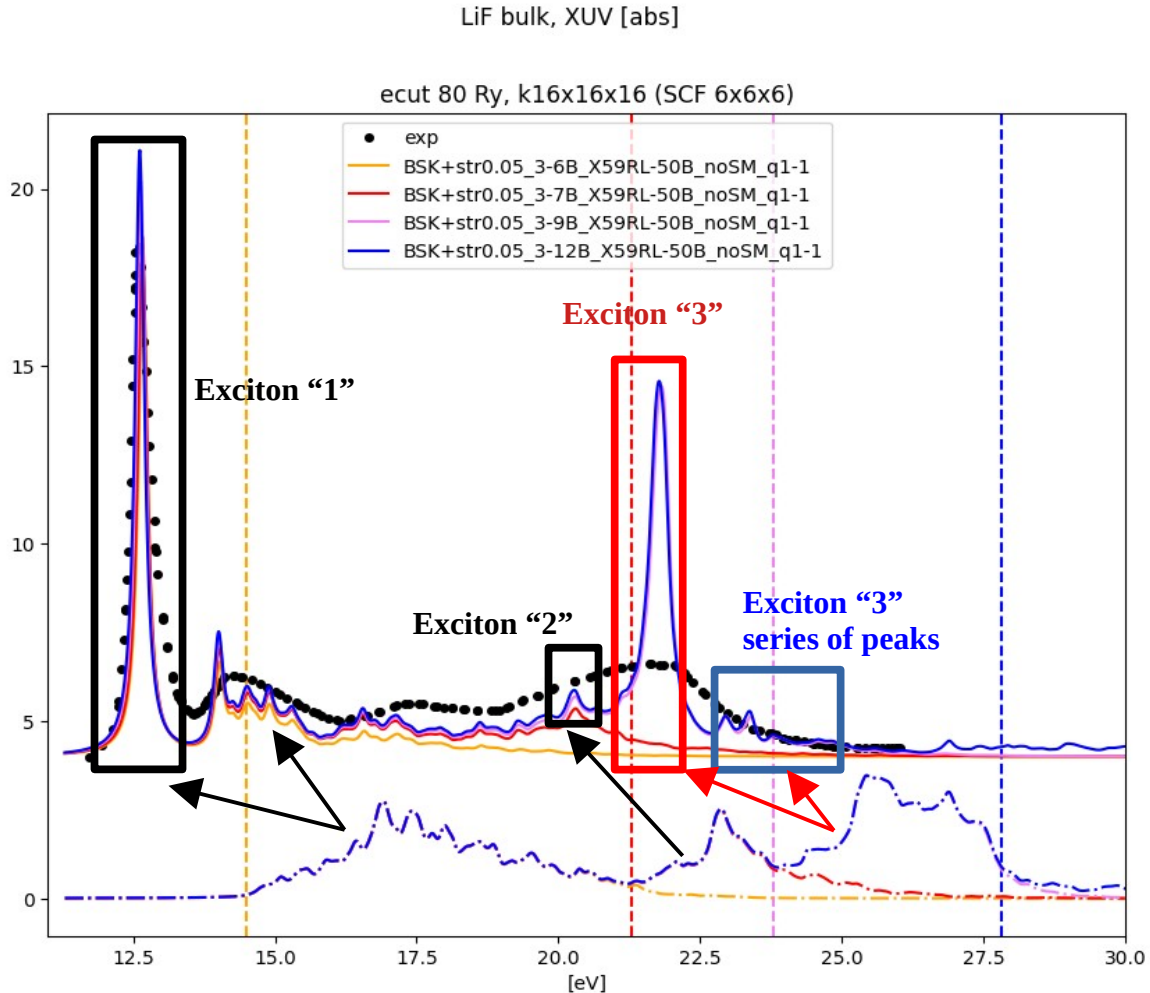


Valence to conduction transitions. 11-30 eV

Hole in bands 3-5; Electrons in bands **6 (orange)**; **6-7 (red)**; **6-9 (pink)**; **6-12 (blue)**

In all plots results are with (continuous line) and without (dot-dashed line) excitonic effects

absorption



a) first excitonic peak at ~ 12.5 eV, with binding 2 eV followed by the Rydberg series entering in the continuum. It is due to transitions from the valence band (states 3-5) to the conduction bands (state 6) around gamma.

b) second excitonic peak at ~ 20.5 eV, with binding 1eV, followed by close laying excitonic peaks. Due to transitions from valence band (states 3-5) to the conduction band (state 7). It is overall weaker compared to the other two excitonic structures

c) third excitonic peak at ~ 22.5 eV, with binding 2.2 eV, followed by the Rydberg series. The peak is stable against the number of bands. From the simulations, it is due to transitions from the valence band (states 3-5) to the conduction bands (states 8-9). To check how it evolves against the k-points sampling and the parameters of the screening. To further study which region in k-space is responsible for it.

It seems that this third series was never computed, but it has been measured here:

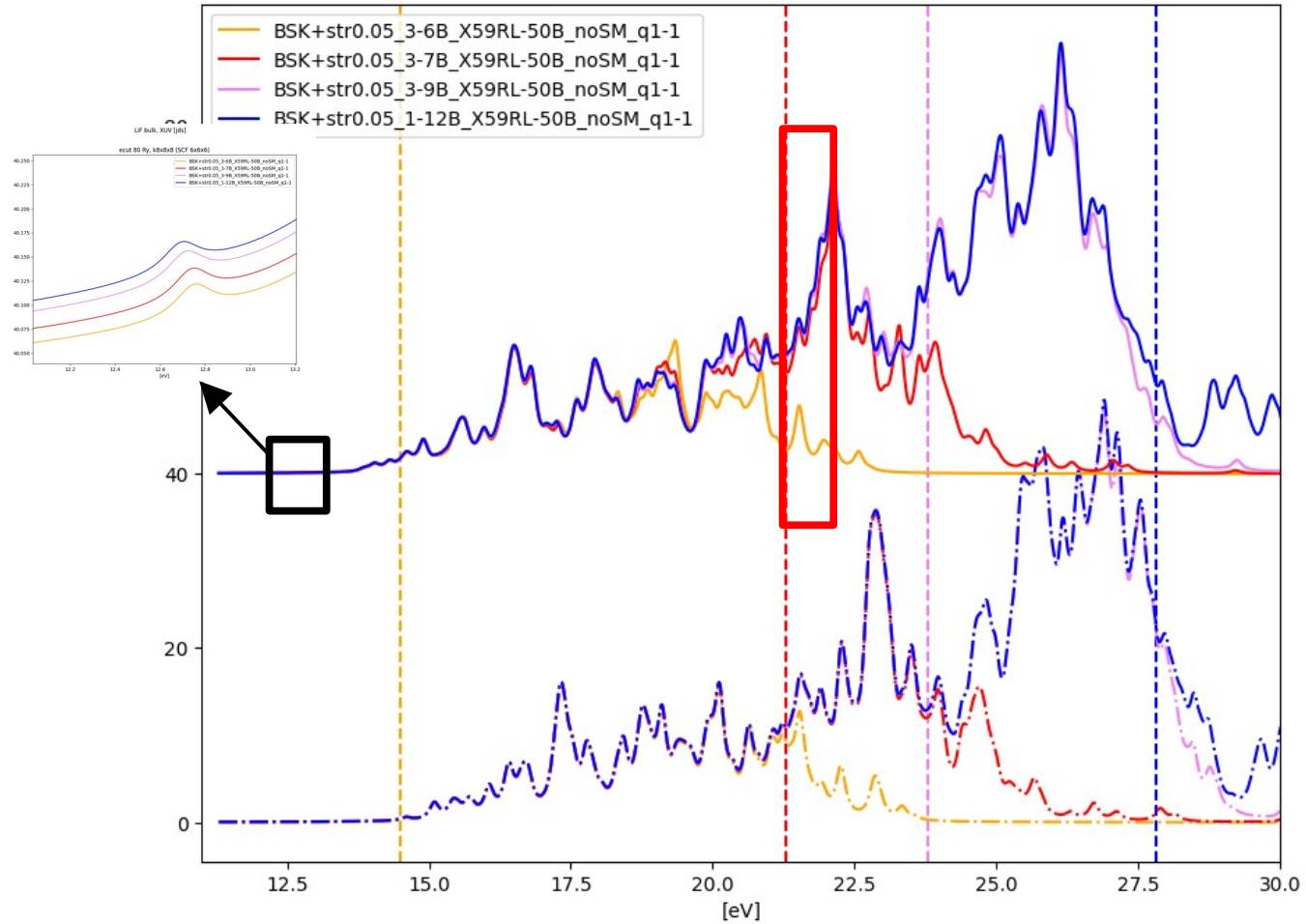
<https://doi.org/10.1103/PhysRevB.13.5530>.

The experimental signal in the extracted imaginary part of the dielectric function around 20-25 eV quite broad, likely because it's in the continuum.

joint/excitonic density of states

LiF bulk, XUV [jds]

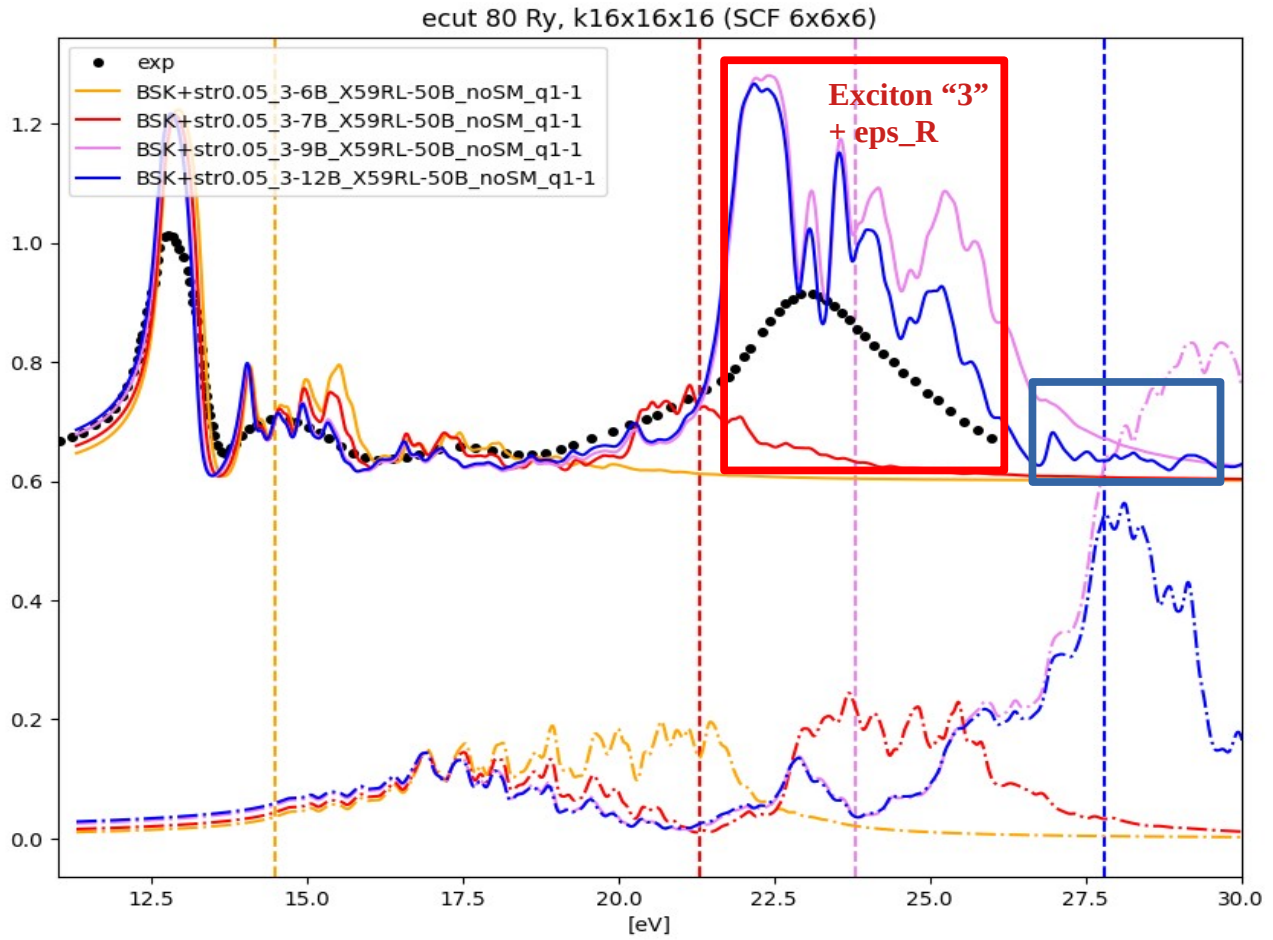
ecut 80 Ry, k8x8x8 (SCF 6x6x6)



While excitonic energy shifts are visible in the DOS as well, there is no re-normalization of the intensity, which is a purely dipolar effect

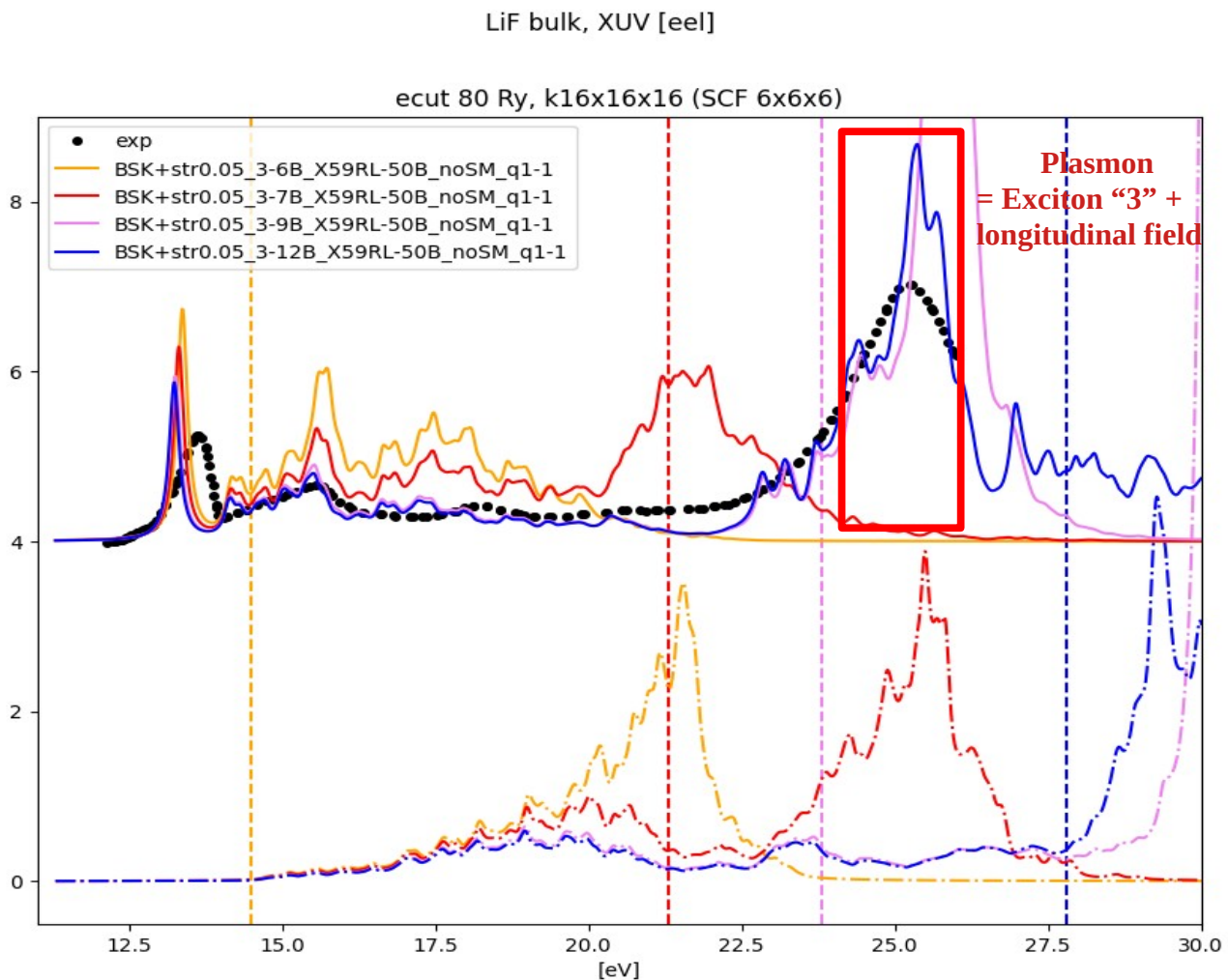
Here $jdos$ with $nscf$ 8x8x8, todo 16x16x16 as other plots.
For impact of k -grids on absorption see last slides

reflectivity



Compared to absorption, there is a broader feature in the reflectivity connected to the exciton "3", which goes from 22 to 25 eV. This effect is due to the real part (see later LiF plasmon). Also the tail of such signal extends up to 30 eV. However this is not converged with the k-mesh 8x8x8

electron energy loss (EEL)



From the previously mentioned publication (<https://doi.org/10.1103/PhysRevB.13.5530>) there is a clear signal in the EELS at ~ 26 eV. Such signal has been studied recently and it is instead due to the plasmon of LiF. See for examples these two papers, first experimental and second theoretical:

- <http://www.pnas.org/cgi/doi/10.1073/pnas.0801623105>
- <http://dx.doi.org/10.1103/PhysRevB.88.155113>

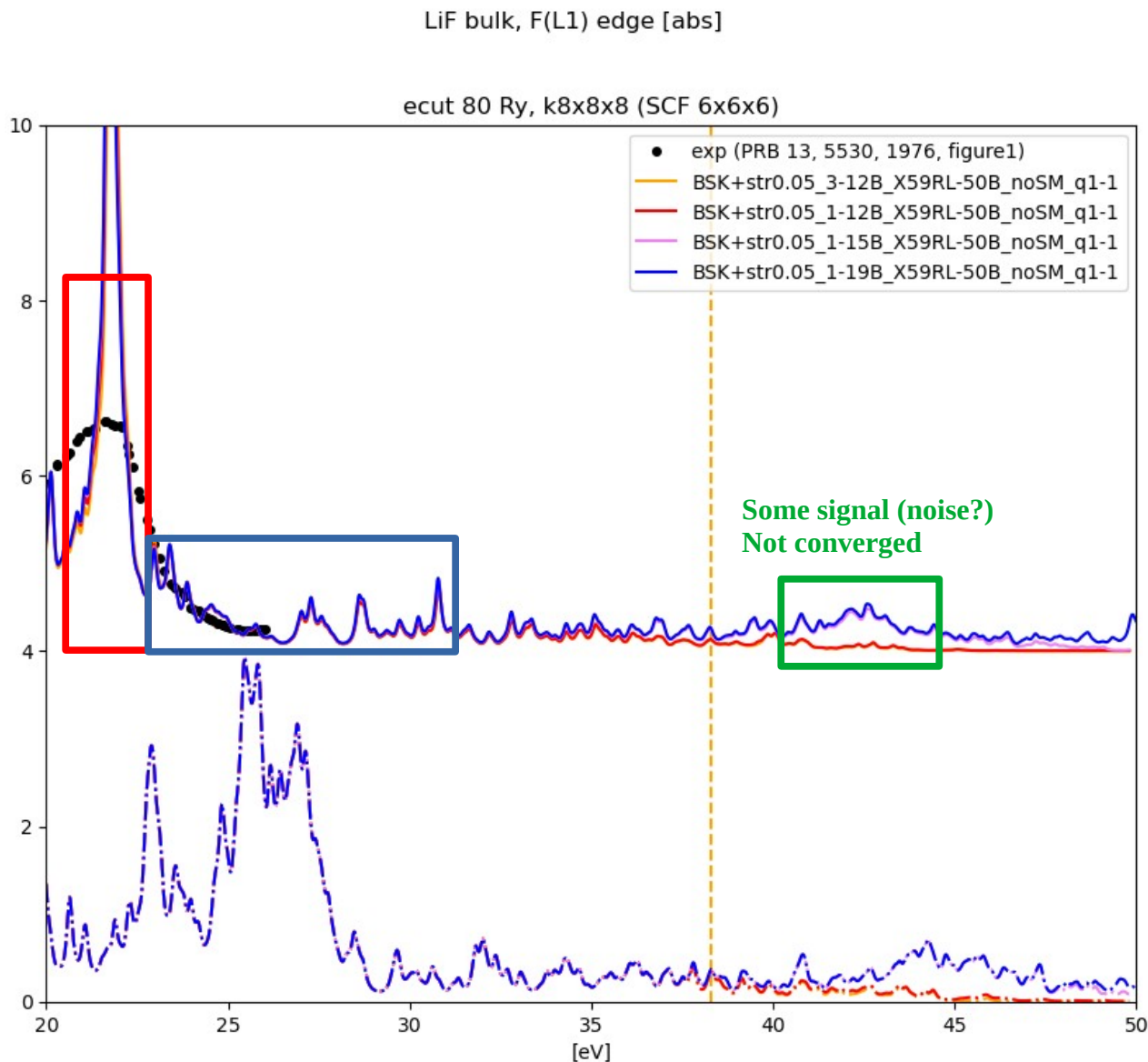
In the above plot of the eel function the plasmon (although not yet converged) is appearing (blue line, peak around . The three previously discussed group of transitions, which give rise to 3 series of excitons (transverse peaks) are affected by the long range Coulomb interaction (which is the source of the LT splitting) giving three set of longitudinal peaks. The lowest energy one remains stable, while instead the second and third series mixes strongly and are likely the source of the plasmon which is starting to appear (blue line). Since the transverse excitations are already converged and stable against the number of bands, this implies that it is the long range exchange interaction which is responsible for the mixing with higher energy transitions.

Here, already at the "IP" level (here "IP eel" includes the long range term of the Coulomb interaction, which enters via the inversion of the dielectric function) the 3 set of transitions are strongly shifted and mixed (see trend from orange to red, pink, blue)

Spectra nearby the F(L1) edge. 20-50 eV

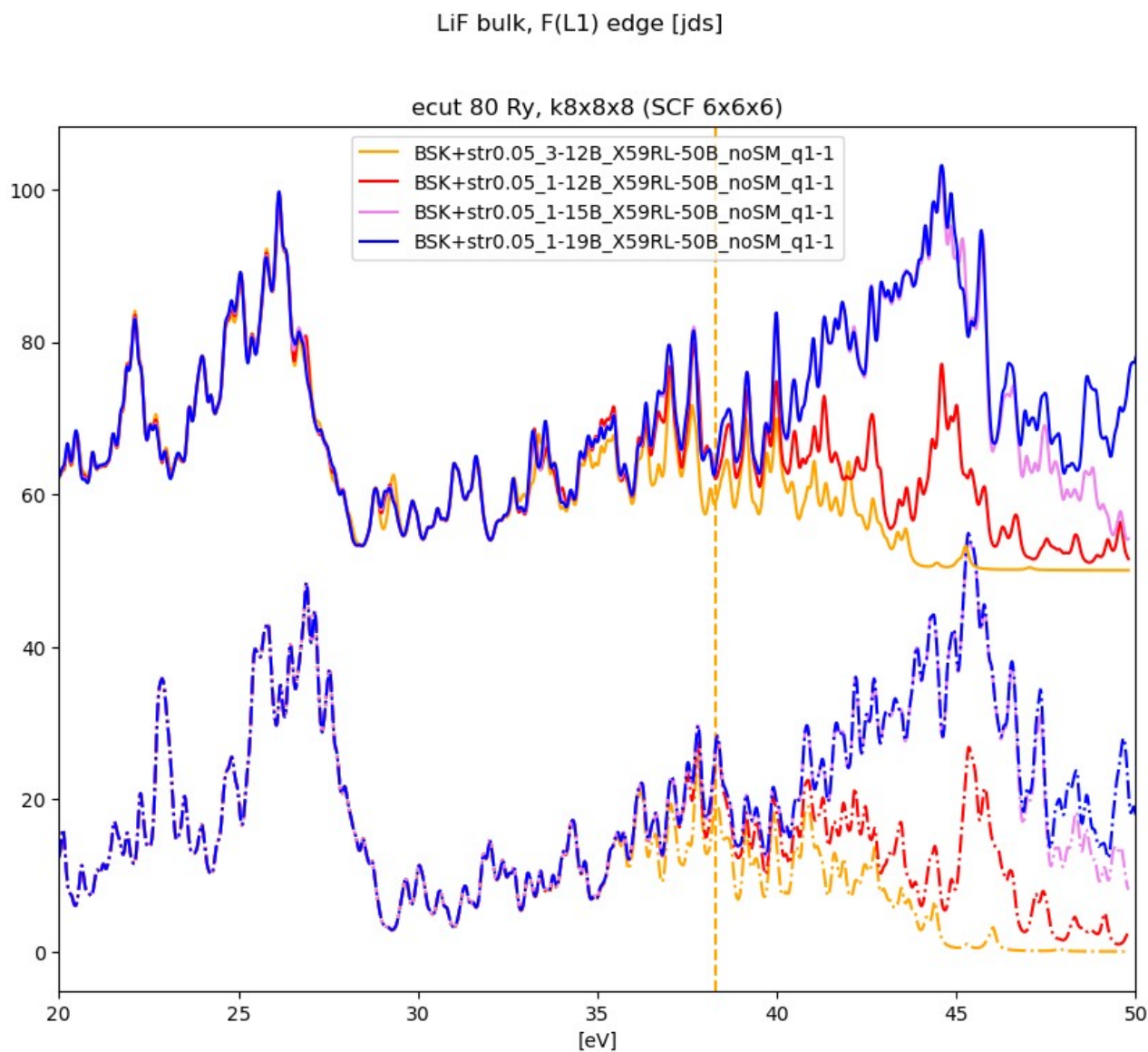
Hole in bands 1-2 (Li 1s and F 2s), and 3-5 (valence);
Elects in bands 6-12 (orange/red); 6-15 (pink), 6-19 (blue).

absorption



The F(L1) edge signal is weak, with almost no difference between with and without it around 45 eV (orange vs red). The low dipole maybe due to the fact that the hole is sitting on F states, while transitions with holes sitting on Li states are brighter (see K edge and spectrum in the UV/XUV region). Such signal is completely covered by the signal (blue - violet) from valence (3-5) to states in the continuum (10-12). The latter gives some signal at around 43 eV. Not really converged with the present k-mesh (8x8x8). Indeed with the finer grid (16x16x16) used for the absorption in the 10-30 eV energy range, this region is in part different. See also the second last slid.

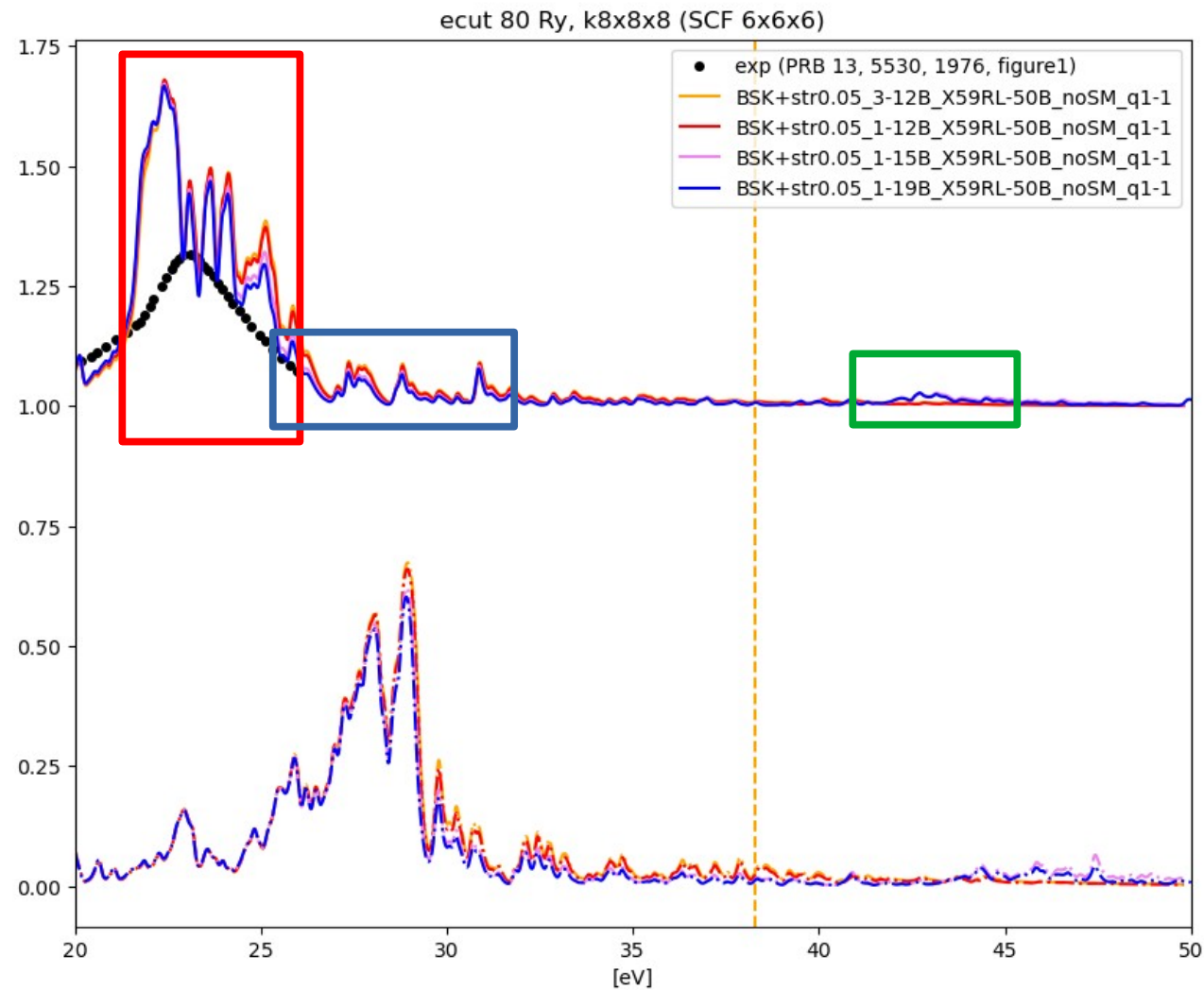
joint/excitonic density of states



The F(L1) edge is clearly visible in the joint/excitonic DOS, difference between with and without it around 45 eV (orange vs red). The maximum in the DOS does not seem to be connected with the signal in the absorption

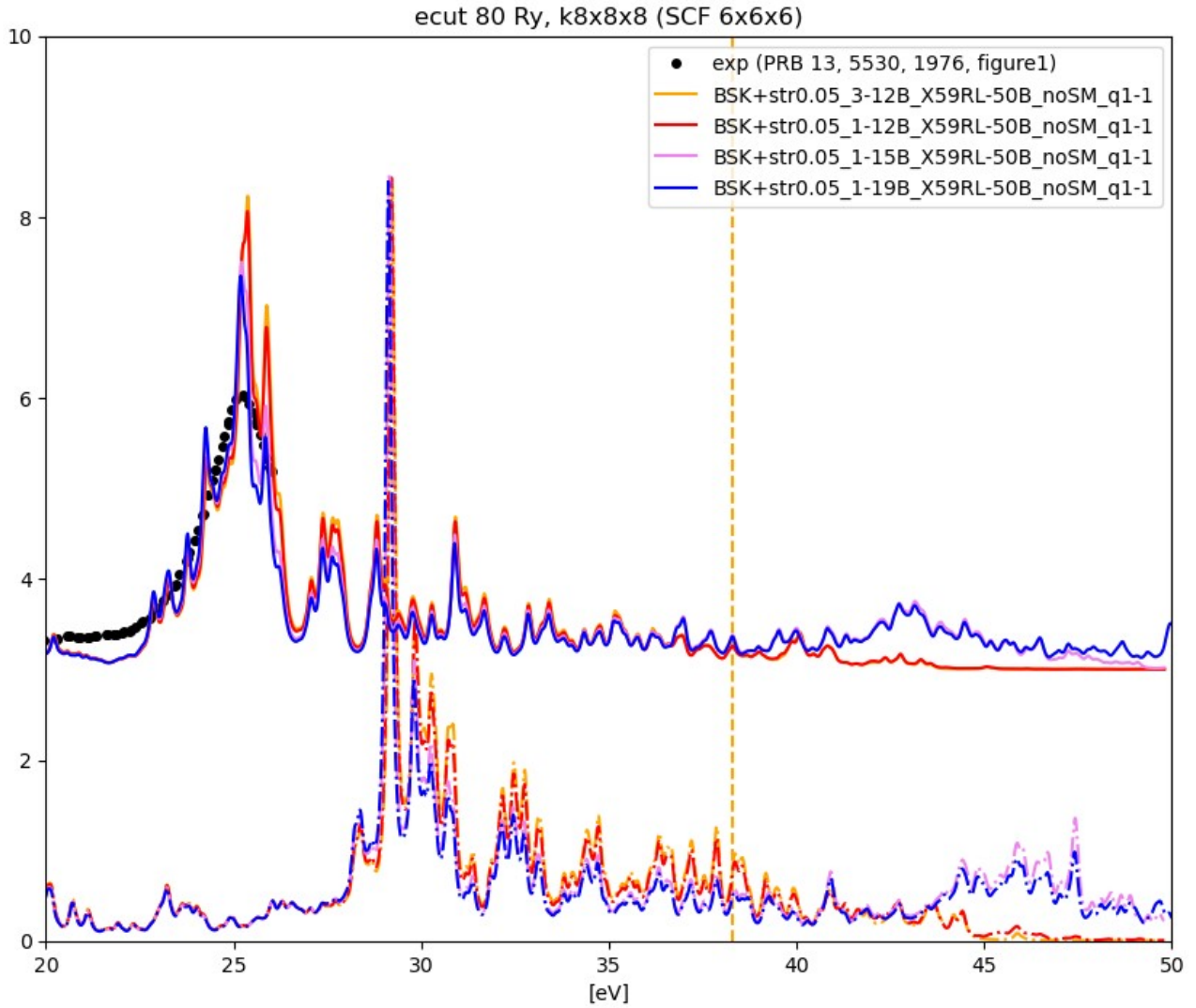
reflectivity

LiF bulk, F(L1) edge [ref]



electron energy loss (EEL)

LiF bulk, F(L1) edge [eel]



Compared to the analysis in the energy range between 12 and 30 eV, here the plasmon is converged w.r.t. the total number of bands used in the simulations.

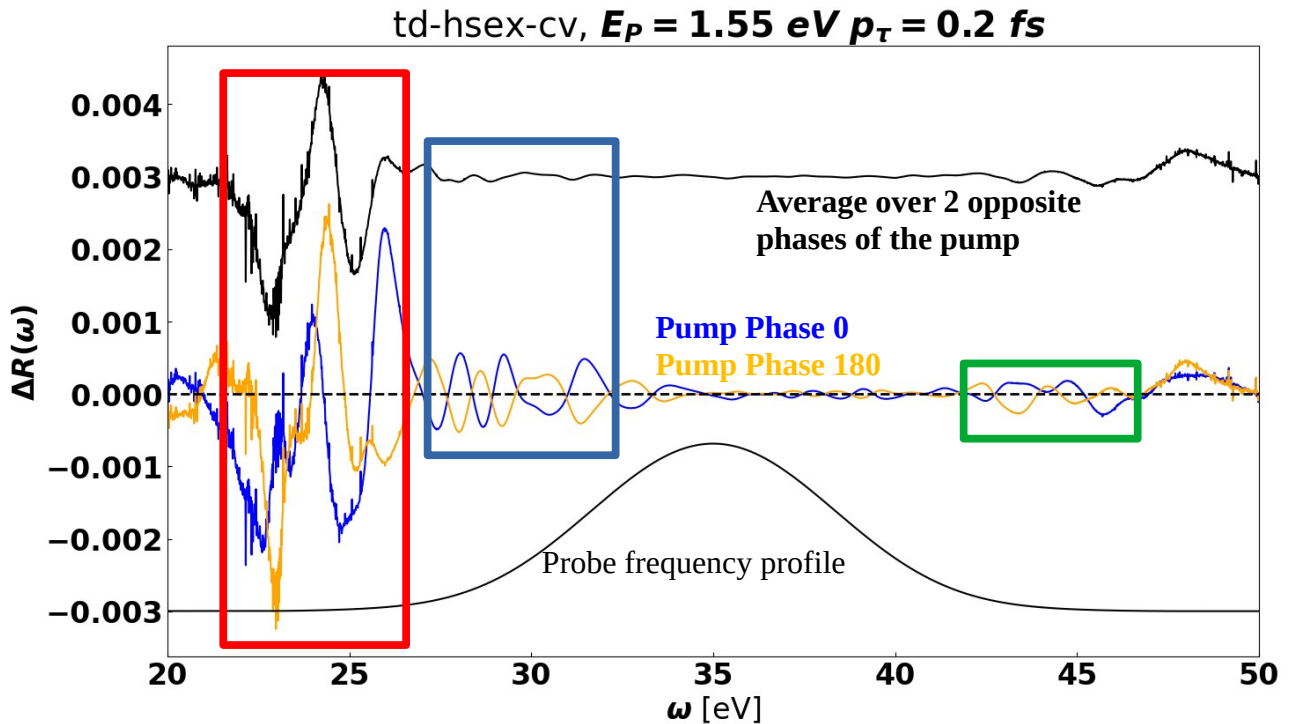
Transient reflectivity (preliminary)

Pump 1.55 eV, Gaussian FWHM ~23.5 fs

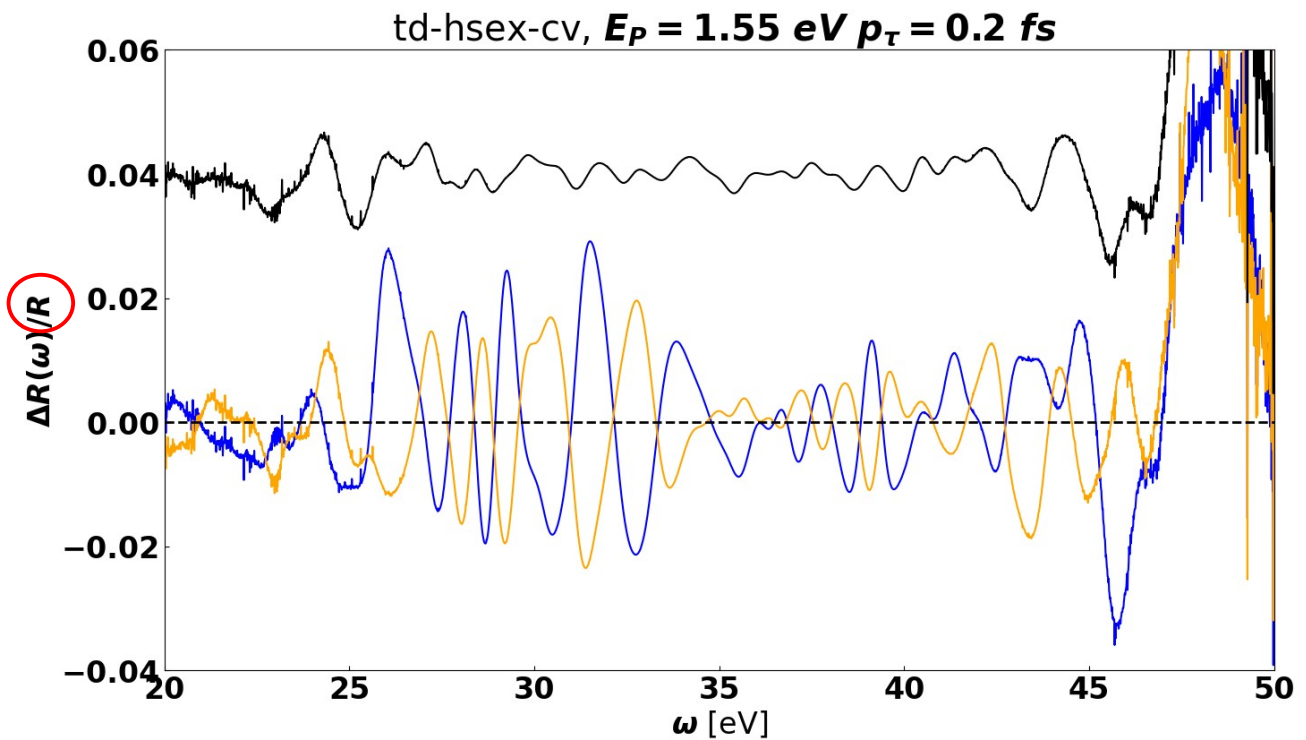
Probe 35 eV, Gaussian. FWHM ~ 470 as

$\tau = -3$ fs, w.r.t. pump maximum

(2 set of simulations, pumps with opposite phases used)



Dividing by the equilibrium spectrum strongly enhances the signal at low intensity

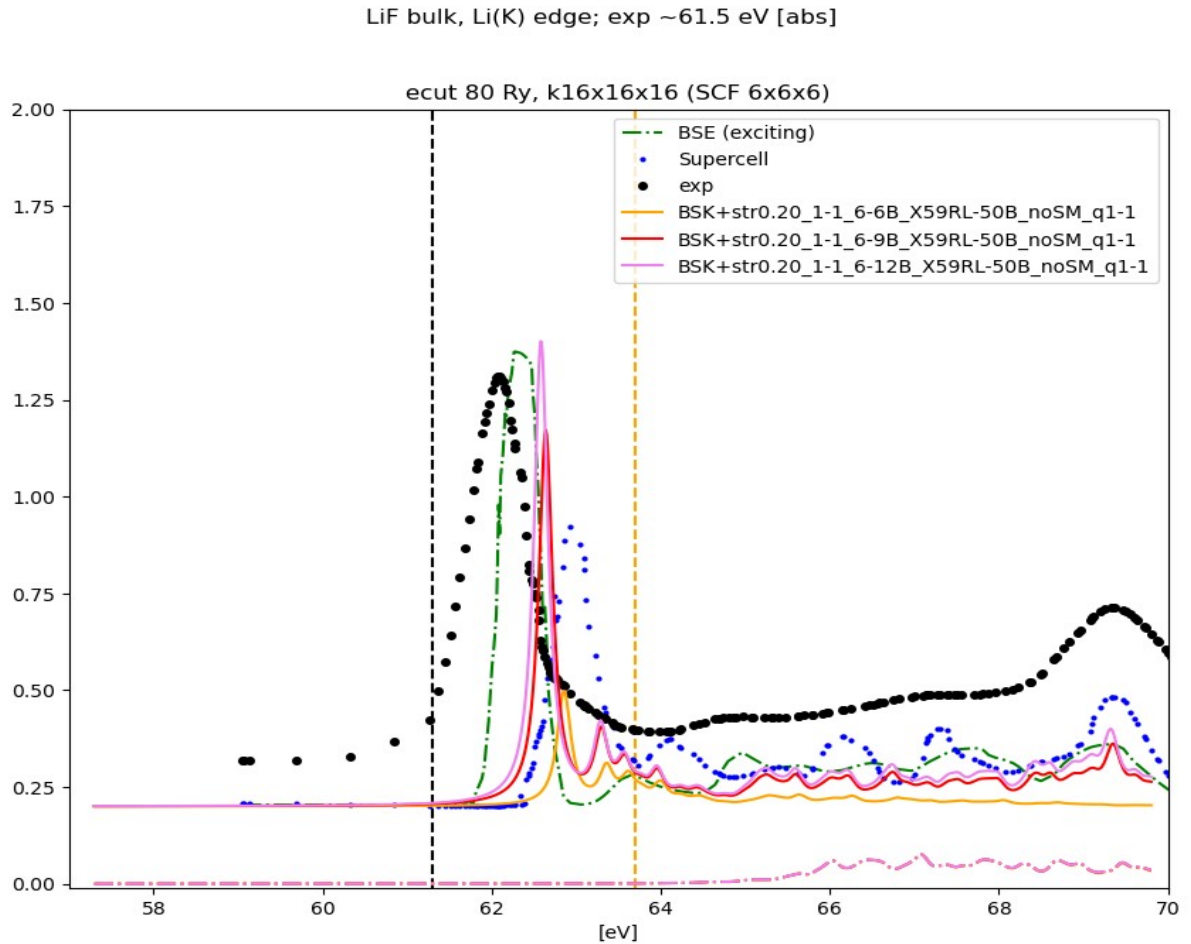


better kpts convergence might be needed

Spectra nearby the Li(K) edge. 57-70 eV

Absorption (kx16), core hole only

Hole in band 1 (Li 1s); Elec. in bands 6 (orange); 6-9 (red); 6-12 (violet).

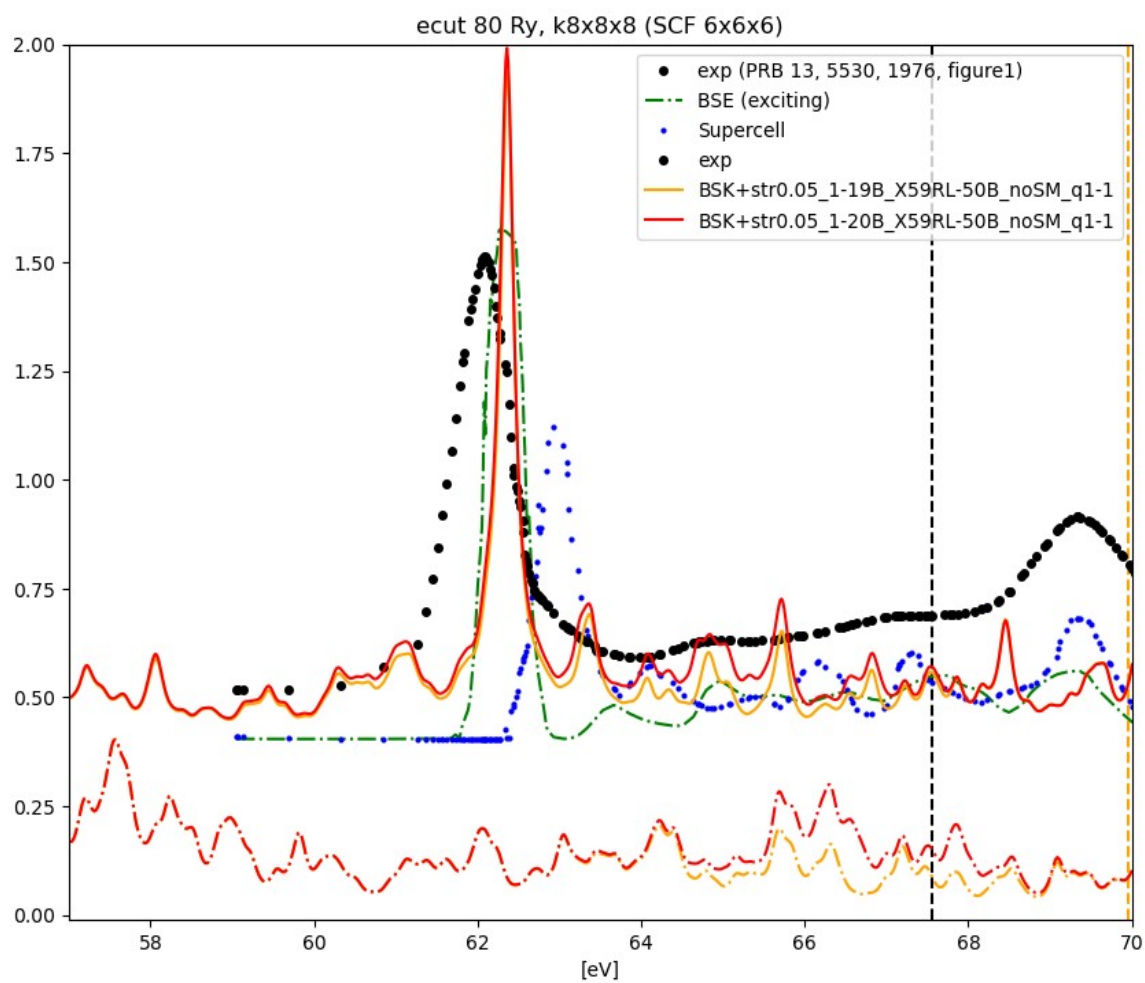


The K edge is quite sharp. The edge is a factor x10 weaker than the absorption of the conduction/valence sector (see later). It is due to transitions from the core hole (state 1) to the conduction (state 6). Convergence due to other states affects the final result. Here the peak position is shifted by about 5 eV.

Absorption (kx8), all holes

All states from 1 to 19/20.

LiF bulk, Li(K) edge; exp ~61.5 eV [abs]



(TODO ?)

reflectivity

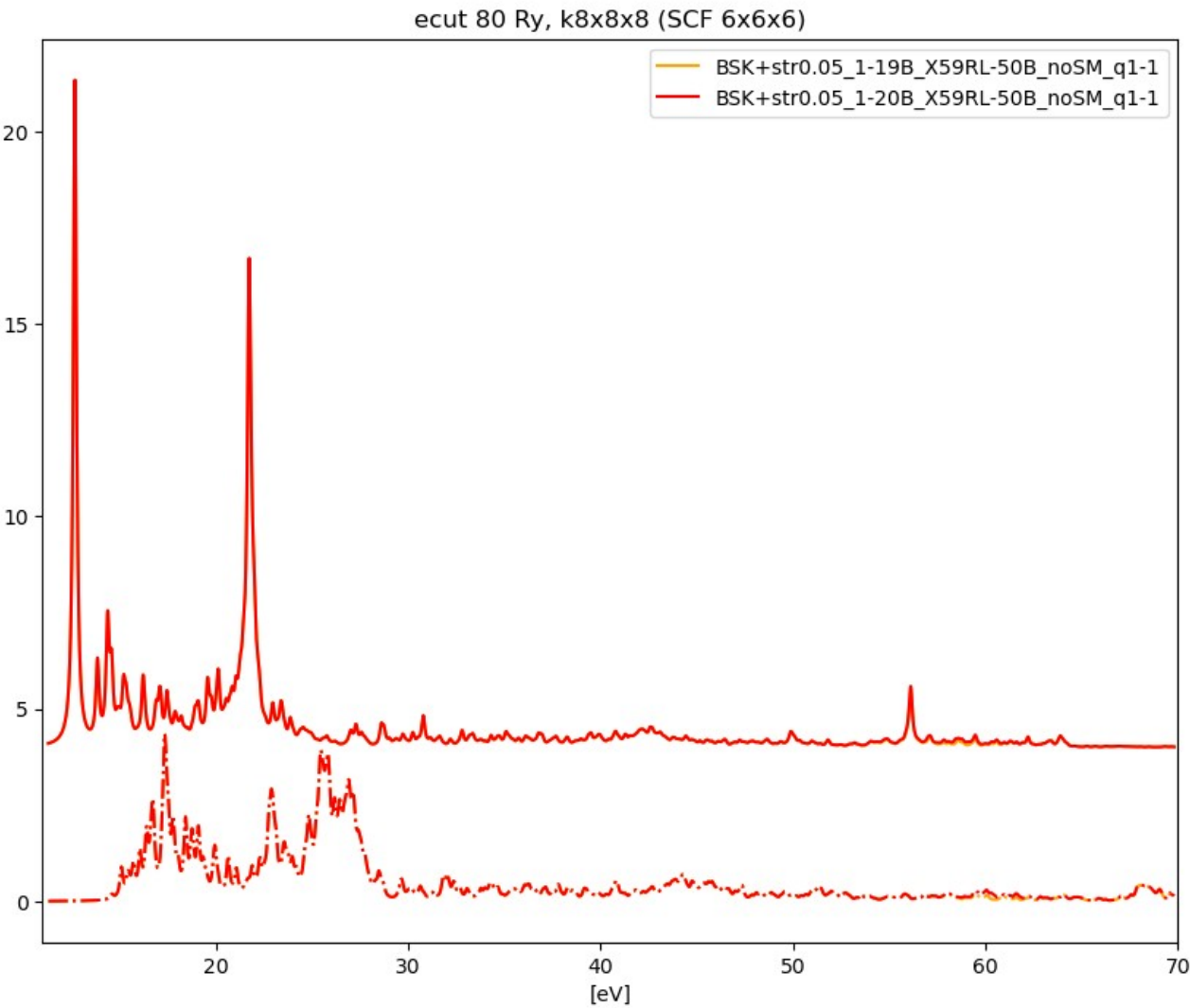
electron energy loss (EEL)

joint/exictonic density of states

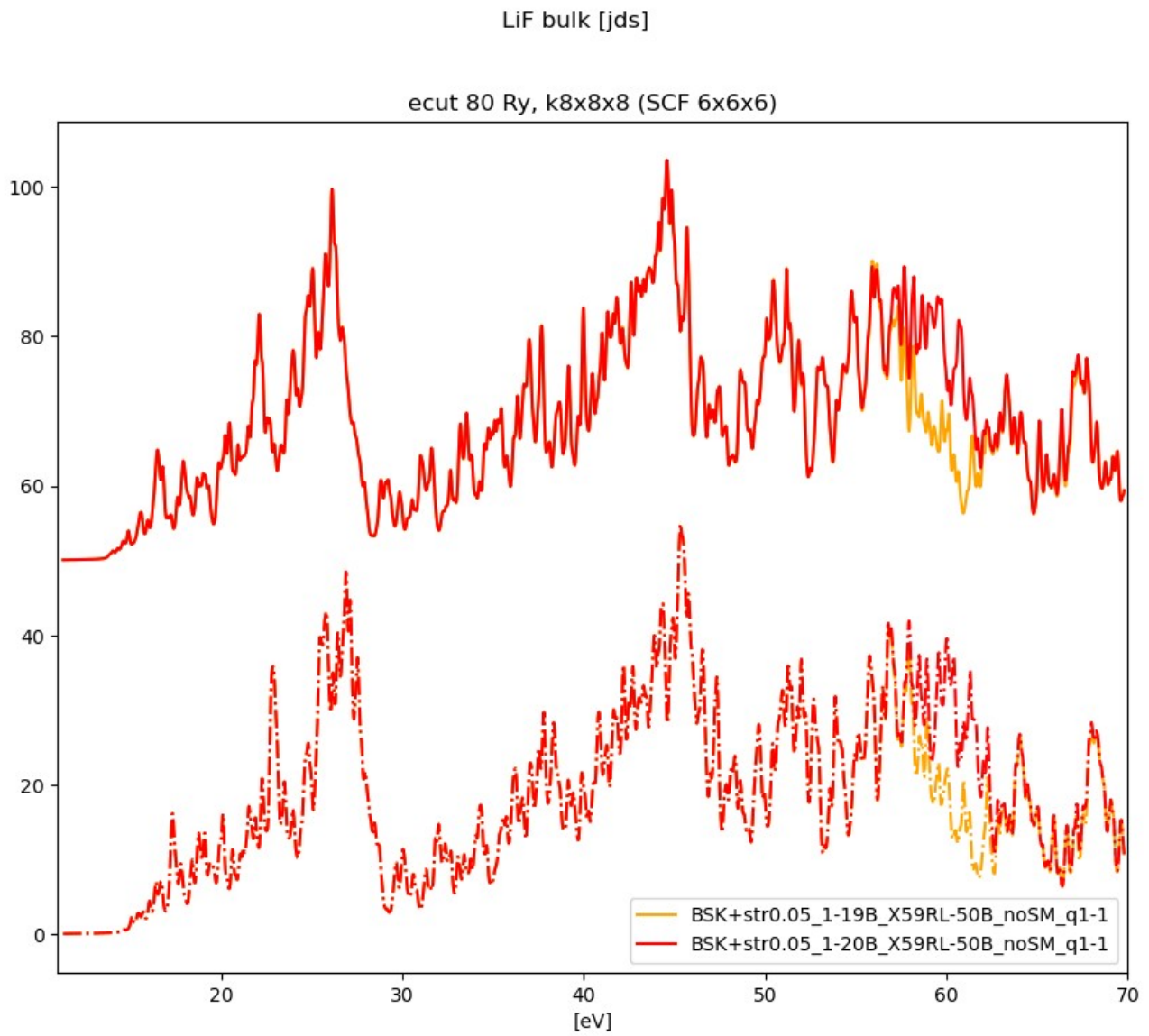
Whole spectrum

Absorption (kx8)

LiF bulk [abs]



joint/exictonic density of states



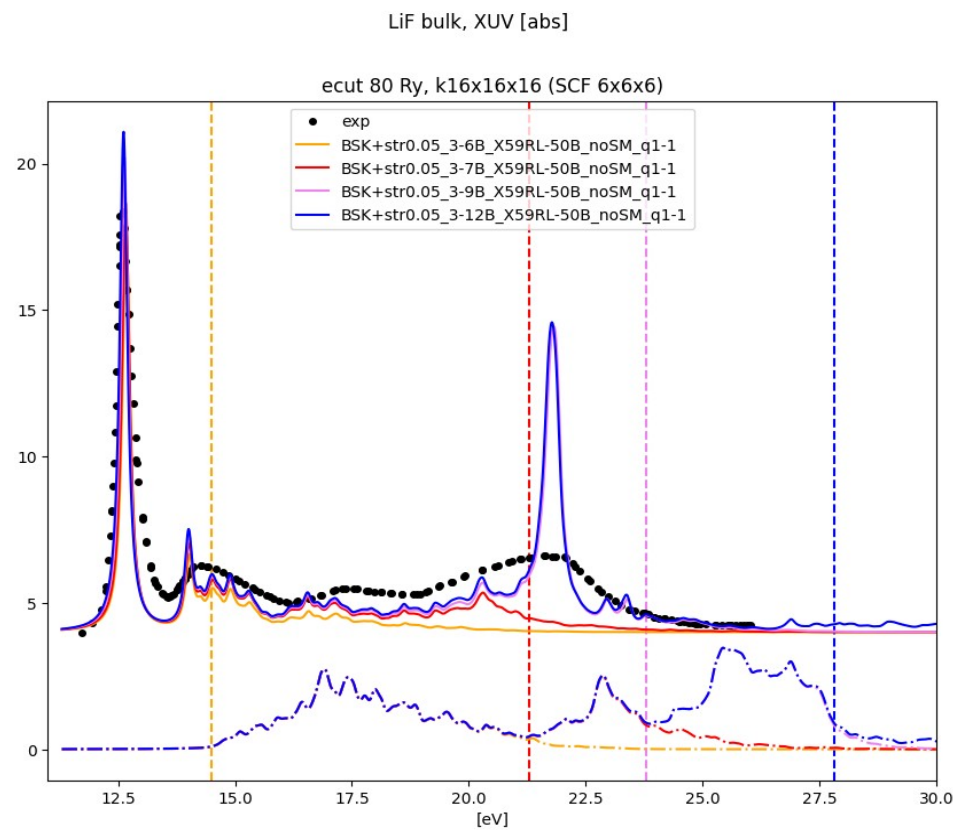
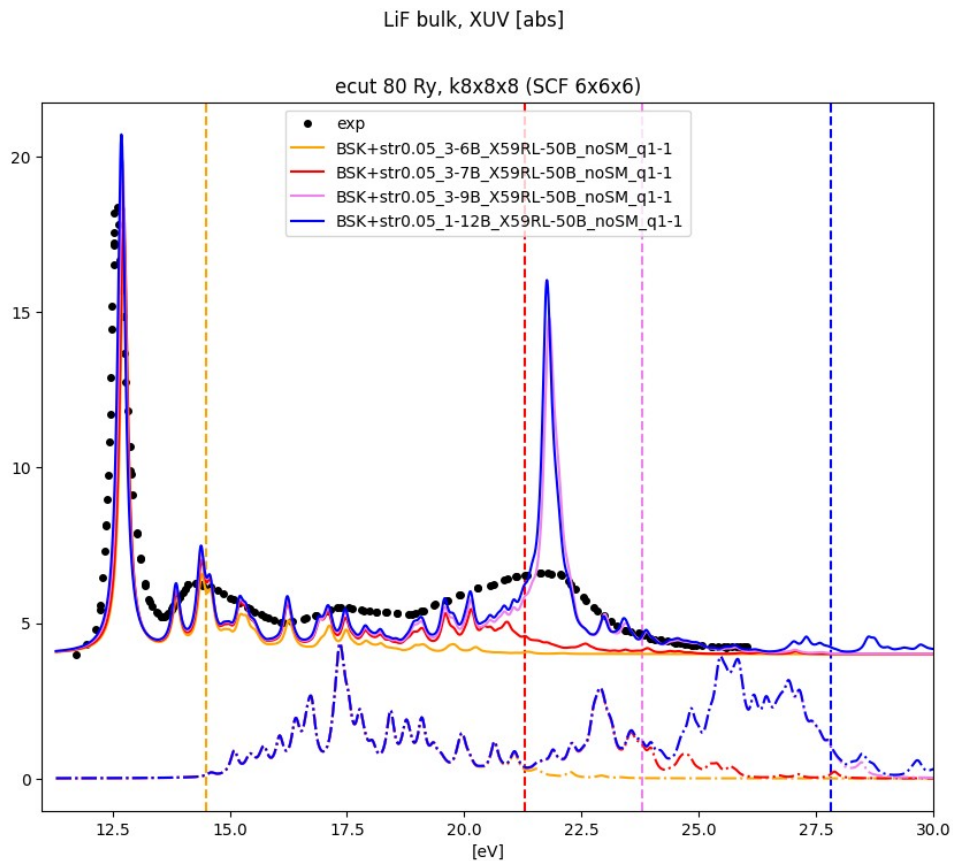
(TODO ?)

reflectivity

electron energy loss (EEL)

MORE

BSE and kpts convergence (kx8 vs kx16)



BSE 2-5B convergence for parameters screening cutoff and nb

The plot below is just to confirm that the parameters used for the screening are ok.

There is a shift of ~ 1 meV by changing the number of bands from 50 to 100 and almost no change if the energy cutoff is increased from 59 to 65 RL

